

Explaining wage variability in the US using OLS models

Introduction

Wages are a function of multiple socioeconomic characteristics including age, gender, experience, education, type of housing, transportation methods etc. The most prevalent variable is education and therefore economies prioritise investment in education in order to improve human capital as a positive relationship is exhibited between years of education and wages. For example, in the US, per pupil public K-12 spending averaged \$16,526 in FY 2023 (Hanson, 2020). Another important variable is work experience which together with education are the key variables of the human-capital earnings OLS model which is investigated in multiple pieces of literature to examine relationships with wages.

An econometric problem arises however when investigating the causal effect of education (amongst alternative variables) on earnings. This is because, the causal effect of schooling on education can vary with other individual characteristics such as ability and therefore can lead to bias in OLS estimates.

This report aims to use 3 OLS regressions to examine the relationship between the socioeconomic characteristics mentioned and hourly wages. Firstly, a simple OLS regression of wages on education will examine the relationship and explanatory power of increasing years of education on wages. The following regressions will add work experience and additional explanatory variables such as gender, tenure etc to see whether the explanatory power of the model increases and understand any other underlying relationships that occur in real life such as the gender wage gap.

Literature review

Education and wages:

As education is associated with higher earnings, the human-capital investment model provides a theoretical motivation for this. Blundell et al (1999) report from the Institute for Fiscal Studies examines returns to education at 3 levels: the individual, the firm, and the economy (Blundell et al., 1999).

The report explains that there are 3 main components of human capital: early ability, qualifications, and knowledge acquired through education, skills, or work experience. Whilst human capital investments involve an initial cost e.g. tuition, reduced wages on training etc, individuals and firms yield future returns. On a national scale, the main conclusion is that education contributes meaningfully to growth (Blundell et al., 1999).

At an individual level, problems arise when estimating the true causal effect of education and training on individual earnings. The main issue is whether higher earnings that are observed for better educated or higher trained workers are caused by their higher education or training or whether individuals with greater earning

capacity and ability choose to acquire more of these. Secondly, if education or training is measured with error, estimates will be too small (Blundell et al., 1999).

The average return for an extra year of education in the UK is between 5%-10% and it is found that formal qualifications matter more than years as well as returns varying by subject, country, and over time. Individual returns are also shaped by ability, family background, and the local environment (Blundell et al., 1999).

Therefore, education is an important explanatory variable to be used in an OLS linear regression such as in this report. However, whilst using education as an explanatory variable for OLS is useful when interpreting causal relationships, OLS education coefficients may not capture returns accurately due to ability, family background, measurement error and other issues.

Econometric concerns:

Card (1999) investigates the causal effect of education on earnings, highlighting how caution must be taken when drawing inferences about the causal effect of schooling as education can vary with other individual characteristics such as family background, and ability (Card, 1999).

His paper looks at heterogeneous returns to schooling and concludes that the average return to education is not much below the estimate from the human-capital earnings function fit by OLS. Studies using instrumental variables (IV) however find that estimated returns to schooling are in fact 20%-40% above OLS estimates (Card, 1999).

The paper highlights the econometric problems with OLS models. Firstly, omitted variable bias (OVB) can arise through the correlation between unobserved ability and education leading to potentially incorrect model coefficients. Measurement error is another issue in education, leading to potential downwards bias in any OLS estimator of the relationship between education and wages. Lastly, family background information such as parent's education level may be used to identify the causal effect of education. However, children's education outcomes are very highly correlated with the characteristics of their parents in particular with parents' education (strong intergenerational correlation in education) (Card, 1999).

Experience and wages:

Although education seems to be the prevalent explanatory variable, work experience should also be included in the OLS regression. Card (1999) uses the human-capital earnings function which decomposes the log of earnings into 3 explanatory variables: an education component, an experience component, and a quadratic experience term. The OLS model is deemed a good fit, explaining 20%-35% of the variation in observed earnings data (Card, 1999).

This motivates the analysis of this report which initially includes education as an explanatory variable in a regression on wages, then goes on to add work experience

in the second regression and finally in the last model includes a multitude of explanatory variables (an important one of which is gender).

Gender wage gap:

In this report, the 3rd regression includes both explanatory variables of education and work experience along with 4 additional variables. Adding additional explanatory variables should lead to a larger R-squared value and explain a higher proportion of the variation in earnings. Gender is one of the additional variables included in 'dummy' variable form. However, it is important to consider the effect of the gender wage gap and why the human-capital model can be limited in this respect.

Blau and Kahn (2016) report highlights that although the gender wage gap has decreased, by 2014, women in full-time jobs earned only 79% of men's annual earnings. The gender wage gap reduced since the 1980's with women in the 90th percentile gaining the least (the glass ceiling pattern), showing how the unexplained gender pay gap declined much more slowly at the top of the wage distribution than at the middle or bottom (Blau and Kahn, 2016).

The gender wage gap reduced from multiple reasons. Women increased their education, the experience gap decreased from 7 years to 1.4 years over 1981-2011, and male advantage in managerial jobs decreased from 12 to 2 points. However, whilst human capital factors previously explained 27% of the gap in 1980, this only explained 8% in 2010. Instead, occupation/industry segregation matter more along with alternative explanations such as psychological traits. An unexplained gap also persists specifically at the top of the distribution (Blau and Kahn, 2016).

Conclusion:

This study models how education singularly explains wage variation, then combining this with work experience in a second model and finally introduces 4 other variables including gender in the 3rd model (full model). These models aim to examine whether education-wage association and female wage differentials persist after adding controls.

Literature suggests that the regression results should highlight positive relationships between education, work experience and wages, whilst also capturing the gender pay gap in the full model. However, causal interpretation should be undertaken with caution due to possible issues arising from omitted variable bias, measurement errors, and family background issues depending on how the data was collected.

Data:

The dataset titled 'wage1' is real data from the 1976 current population survey in the US. The dataset is accessed through the RDatasets repository maintained by Vincent Arel-Bundock. The link to the dataset can be found here:

<https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/wage1.csv>. It is an established econometrics dataset for applied OLS designed around wage, education, experience and tenure. Additionally, the dataset is specifically used in Wooldridge's Introductory Econometrics and is therefore appropriate for an OLS project.

The dataset contains 526 observations for wages with 23 potential explanatory variables. However, in this report only 6 explanatory variables will be modelled and are summarized in the table below. There are no missing values that need to be accounted for. Additionally, it is important to note that the 'female' and 'married' explanatory variables are dummy variables. The data shows that there are 274 males and 252 females, with 320 of individuals that are married and 206 that are not married.

Variable	Code name	Variable type	Definition
Wage	wage	Integer	Hourly wage (\$)
Education	educ	Integer	Years of education
Experience	exper	Integer	Years of potential work experience
Tenure	tenure	Integer	Years with current employer
Gender	female	Categorical	Dummy variable (1 if female, 0 if male)
Marital Status	married	Categorical	Dummy variable (1 if married, 0 if not married)
Number of dependents	numdep	Integer	Number of dependents in the household

In this report, the relationship between explanatory variables educ, exper, tenure, female, married and numdep and wages is investigated.

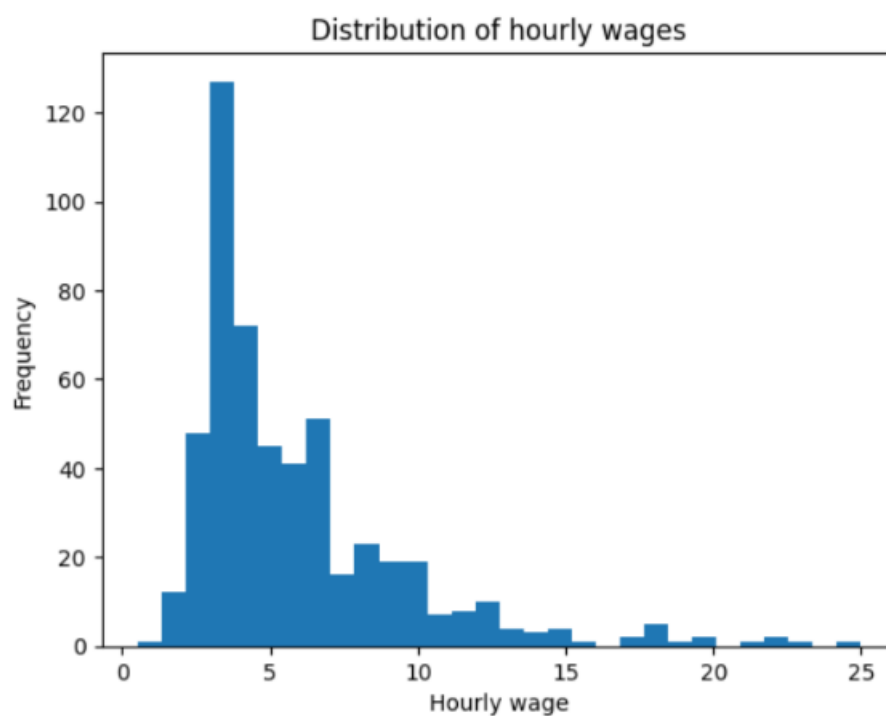
EDA:

Descriptive statistics:

	wage	educ	exper	tenure	female	married	numdep
count	526.000000	526.000000	526.000000	526.000000	526.000000	526.000000	526.000000
mean	5.896103	12.562738	17.01711	5.104563	0.479087	0.608365	1.043726
std	3.693086	2.769022	13.57216	7.224462	0.500038	0.488580	1.261891
min	0.530000	0.000000	1.00000	0.000000	0.000000	0.000000	0.000000
25%	3.330000	12.000000	5.00000	0.000000	0.000000	0.000000	0.000000
50%	4.650000	12.000000	13.50000	2.000000	0.000000	1.000000	1.000000
75%	6.880000	14.000000	26.00000	7.000000	1.000000	1.000000	2.000000
max	24.980000	18.000000	51.00000	44.000000	1.000000	1.000000	6.000000

These initial descriptive statistics examine wage trends in 1976. The average wage was \$5.90 per hour. Average years of education is 12, and average years of work experience is 17. The maximum wage reached \$25 and the minimum was around \$0.50.

Distribution of hourly wages:



The histogram above provides a visual representation for the distribution of hourly wages in the sample. Most hourly wages are concentrated around \$3-\$7 per hour

with only a small number earning substantially more. The upper tail is large with some individuals earning up to \$25 per hour however this is only a very small proportion.

Methodology:

The OLS specification for the model is as follows. The predicted variable is $wage_i$. In model 3, π_0 is the intercept term and $\pi_1, \dots, \pi_6$ are the coefficients on the explanatory variables which represent conditional associations not necessarily the causal effects. Finally, γ_i is the error term which contains any additional information that may explain the variation in $wage_i$ which is not captured by the model.

Regression 1:

$$wage_i = \beta_0 + B_1educ_i + \varepsilon_i$$

Regression 2:

$$wage_i = \alpha_0 + \alpha_1educ_i + \alpha_2exper_i + \mu_i$$

Regression 3:

$$wage_i = \pi_0 + \pi_1educ_i + \pi_2exper_i + \pi_3tenure_i + \pi_4female_i + \pi_5married_i + \pi_6numdep_i + \gamma_i$$

The primary aim of this report is to explain the relationship between the characteristics of individuals and the hourly wage that they receive. This will be evaluated in the form of 3 regressions. Firstly, in 'Regression 1', wage is regressed on education and a constant. Secondly, in 'Regression 2', wage is regressed on education, work experience, and a constant. Lastly, in 'Regression 3', the full model regresses wage on multiple variables including education, work experience, tenure, gender, marital status, number of dependents, and a constant.

After running each model in Python, the effect of each explanatory variable on income can be computed by interpreting coefficients, evaluating the statistical significance, and analysing the R squared coefficient.

The three models are estimated sequentially, assessing how the relationship between education and wages changes with the introduction of additional controls. Regression 1 shows the unconditional association between education and wages, and Regression 2 examines whether this association remains after controlling for potential work experience. Finally in Regression 3, further controls for tenure, gender, marital status and number of dependents are introduced. By comparing the coefficients and R^2 values across these models, insight is provided into how the initial education-wage relationship changes as additional explanatory variables are introduced.

Results

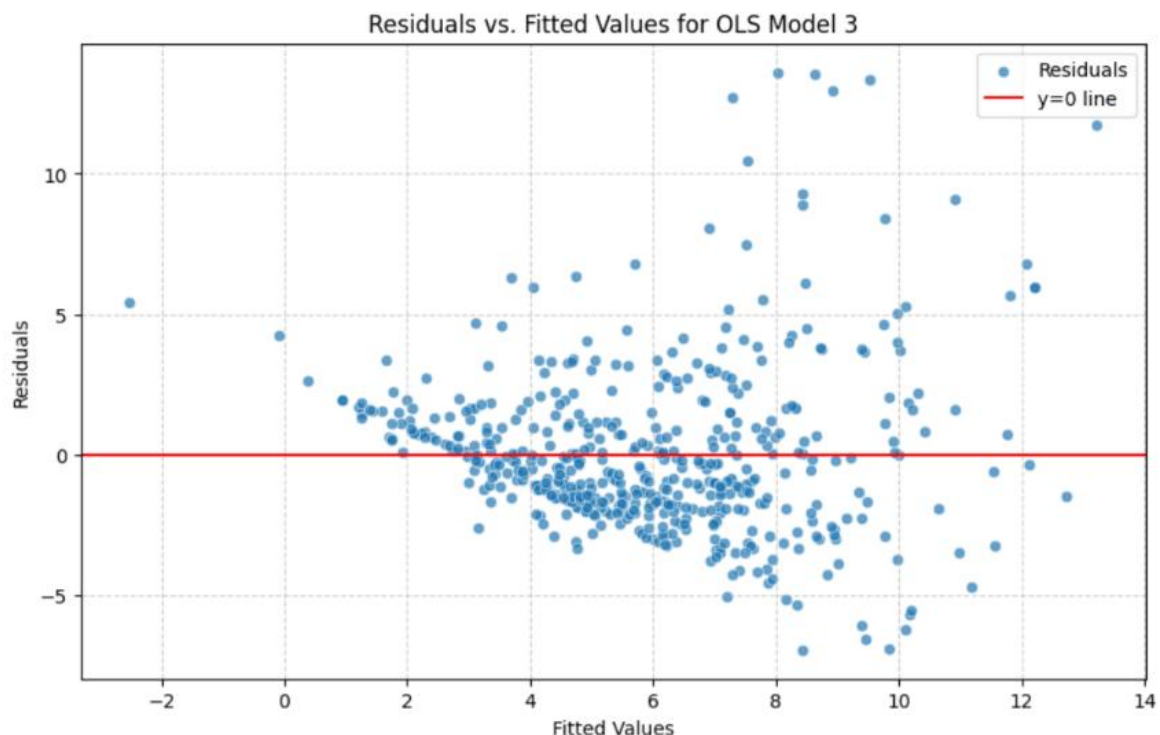
Testing for heteroskedasticity:

```
--- Breusch-Pagan Test for Model 1 ---  
{'Lagrange multiplier statistic': np.float64(15.305628705020107), 'p-value': np.float64(9.144363288378174e-05),  
  
--- Breusch-Pagan Test for Model 2 ---  
{'Lagrange multiplier statistic': np.float64(31.847282579768454), 'p-value': np.float64(1.2146480463878308e-07),  
  
--- Breusch-Pagan Test for Model 3 ---  
{'Lagrange multiplier statistic': np.float64(43.41522476314023), 'p-value': np.float64(9.652906758605745e-08),
```

After all 3 regressions were initially computed, a Breusch-Pagan test was undertaken in order to test for heteroskedasticity. The test results show that all 3 models have p-values much smaller than 0.05, indicating the presence of heteroskedasticity in the regressions. Therefore model adjustments have been made to incorporate heteroskedasticity robust standard errors (HC1 standard errors) and the regressions were performed again.

This is important because whilst heteroskedasticity doesn't make coefficient estimates biased, usual OLS standard errors however become unreliable. Using HC1 robust standard errors therefore allows statistical inference to remain valid under heteroskedasticity, without changing the OLS coefficient estimates.

Regression line for full model:



The plot of residuals vs fitted values for the complete model regression (model 3) visually aids in the detection of heteroskedasticity. Under homoskedasticity, the variance of residuals is constant across all fitted values and should be randomly scattered around 0. A 'funnel' or 'cone' shape which can be seen above indicates heteroskedasticity.

As the spread of residuals changes with the fitted values, this is consistent with the Breusch-Pagan test result.

Testing for multicollinearity:

The VIF test diagnoses multicollinearity via the following:

VIF Score:	Interpretation
<5	Small concern
5-10	Moderate concern
>10	Severe multicollinearity

	feature	VIF
1	exper	1.670458
2	tenure	1.407412
0	educ	1.250892
4	married	1.249400
5	numdep	1.132517
3	female	1.073712

All VIF values are below 5, suggesting that multicollinearity is unlikely to be a serious concern in the full model.

Regression 1 Results:

Regression 1:

$$wage_i = \beta_0 + B_1educ_i + \varepsilon_i$$

```
--- New Regression 1 (HC1 Robust SEs): wage ~ educ ---
                        OLS Regression Results
=====
Dep. Variable:          wage      R-squared:                0.165
Model:                  OLS      Adj. R-squared:            0.163
Method:                 Least Squares      F-statistic:         78.09
Date:                   Mon, 31 Aug 2026    Prob (F-statistic):    1.49e-17
Time:                   16:43:38           Log-Likelihood:       -1385.7
No. Observations:      526              AIC:                 2775.
Df Residuals:          524              BIC:                 2784.
Df Model:              1
Covariance Type:       HC1
=====
                        coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept             -0.9049        0.725      -1.247      0.212      -2.327      0.517
educ                   0.5414        0.061       8.837      0.000       0.421      0.661
=====
Omnibus:               212.554      Durbin-Watson:         1.824
Prob(Omnibus):         0.000      Jarque-Bera (JB):      807.843
Skew:                  1.861      Prob(JB):              3.79e-176
Kurtosis:              7.797      Cond. No.              60.2
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

The first model shows an OLS regression of wage on education and a constant. The output yields a few important results. Firstly, this R^2 statistic is 0.165. This can be interpreted as the fact that 16.5% of variation in wages is explained by years of education whilst the remaining 83.5% is left unexplained by the model and included in the error term.

Secondly, the p-value for education tests the null that education is equal to 0 (no effect on wages) against the alternative that education doesn't equal to 0 (has an effect on wages). As this value is below 0.05, education is statistically significant at the 5% level, providing evidence of a significant association between education and hourly wages. As the coefficient on education is 0.5414, this suggests that an additional year of education is associated with an increase in hourly wage by approximately \$0.5414 for the average individual in this sample.

Regression 2 results:

Regression 2:

$$wage_i = \alpha_0 + \alpha_1 educ_i + \alpha_2 exper_i + \mu_i$$

```
--- New Regression 2 (HC1 Robust SEs): wage ~ educ + exper ---
                        OLS Regression Results
=====
Dep. Variable:          wage      R-squared:                0.225
Model:                  OLS      Adj. R-squared:             0.222
Method:                 Least Squares      F-statistic:         50.32
Date:                   Mon, 31 Aug 2026    Prob (F-statistic):      1.03e-20
Time:                   16:43:38           Log-Likelihood:         -1366.0
No. Observations:       526              AIC:                  2738.
Df Residuals:           523              BIC:                  2751.
Df Model:                2
Covariance Type:        HC1
=====
                        coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept             -3.3905         0.865     -3.920     0.000     -5.086     -1.695
educ                   0.6443         0.065      9.883     0.000      0.517      0.772
exper                  0.0701         0.011      6.376     0.000      0.049      0.092
=====
Omnibus:               196.872      Durbin-Watson:          1.820
Prob(Omnibus):          0.000      Jarque-Bera (JB):        727.160
Skew:                   1.715      Prob(JB):                1.26e-158
Kurtosis:               7.627      Cond. No.                130.
=====

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

Regression 2 builds upon Regression 1 by adding the work experience as an additional explanatory variable. The main results from regression 2 show that adding work experience increases the R^2 statistic to 0.225. Now 22.5% of the variance in wages can be explained through education and work experience which is an increase of 6 percentage points from model 1. Equally, this is the same as saying that the model's explanatory power increases from 16.5% to 22.5%.

Education and work experience are both statistically significant as both p-values are 0 and therefore the null hypothesis that these variables don't affect wages can be rejected.

Interestingly in this regression, an extra year of education increases hourly wage by \$0.64 (holding work experience constant) and an extra year of work experience increases hourly wage by \$0.07 (holding education constant).

Regression 3 Results:

Regression 3:

$$wage_i = \pi_0 + \pi_1 educ_i + \pi_2 exper_i + \pi_3 tenure_i + \pi_4 female_i + \pi_5 married_i + \pi_6 numdep_i + \gamma_i$$

--- New Regression 3 (HC1 Robust SEs): wage ~ educ + exper + tenure + female + married + numdep ---

OLS Regression Results						
=====						
Dep. Variable:	wage	R-squared:	0.370			
Model:	OLS	Adj. R-squared:	0.363			
Method:	Least Squares	F-statistic:	31.92			
Date:	Mon, 31 Aug 2026	Prob (F-statistic):	1.01e-32			
Time:	16:46:32	Log-Likelihood:	-1311.4			
No. Observations:	526	AIC:	2637.			
Df Residuals:	519	BIC:	2667.			
Df Model:	6					
Covariance Type:	HC1					
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	-2.0008	0.917	-2.183	0.029	-3.798	-0.204
educ	0.5753	0.067	8.581	0.000	0.444	0.707
exper	0.0219	0.010	2.114	0.035	0.002	0.042
tenure	0.1382	0.028	4.975	0.000	0.084	0.193
female	-1.7575	0.251	-6.995	0.000	-2.250	-1.265
married	0.4655	0.257	1.814	0.070	-0.037	0.968
numdep	0.1443	0.095	1.517	0.129	-0.042	0.331
=====						
Omnibus:	187.184	Durbin-Watson:	1.788			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	715.208			
Skew:	1.605	Prob(JB):	4.95e-156			
Kurtosis:	7.725	Cond. No.	151.			
=====						

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Under the full model in Regression 3, wage is regressed on education and work experience along with the additional explanatory variables:

- Tenure
- Gender
- Marital status
- Number of dependents

Results show that the estimated constant is -2.01 which represents the predicted hourly wage when all explanatory variables are equal to 0 (the individual is a male and not married). However, this interpretation has little economic relevance as 0 years of education, experience, tenure etc doesn't represent a realistic individual in the sample.

The R^2 value has increased to 0.370 when adding these extra explanatory variables. This means that now 37% of the variance in wages can be explained by the model. However, the majority still remains unexplained by the model.

The p-values for the education, work experience, tenure, and female explanatory variables lie below 0.05 and therefore these variables are statistically significant. However, the p-values for numdep and married are 0.1290 and 0.07 respectively. As these exceed 0.05, there is insufficient evidence to conclude that the number of dependents and being married is statistically associated with hourly wages, conditional on the other variables in the model.

The coefficients suggest that being married and spending an additional year with a current employer has a positive impact on wage. Holding the other explanatory variables constant, the female coefficient indicates that females have an estimated hourly wage \$1.76 lower than males, on average. Additionally, holding the other explanatory variables constant, married individuals have an estimated hourly wage of \$0.46 higher on average than unmarried individuals.

Discussion:

Regression 1:

The results from regression 1 show that the education coefficient is positive and significant. This means that additional years of education are associated with higher hourly wages and explain the variation in wages. Literature confirms this should be the case under the Blundell et al (1999) report where the average return for an extra year of education in the UK is between 5%-10% (Blundell et al., 1999).

Regression 2:

The results from regression 2 show that work experience as an explanatory variable is also positive and significant. Therefore, adding work experience as an explanatory variable on top of education means that more of the variation in wages can be explained. However, wages increase less to an additional year of experience than an additional year of education. The literature on returns to experience confirm this relationship to be true as the OLS human-capital model fit investigated by Card (1999) (which captures experience up to a polynomial of order 2) is a good model fit (Card, 1999).

Regression 3

The results show that when adding tenure, gender, marital status, and number of dependents in regression 3 leads to an R^2 value of 0.370 which means that 37% of the variation in wages can be explained by these explanatory variables. Card (1999)

finds that OLS explains 20%-35% of the variation in wages using the human capital model. This report finds that OLS explains 37% which is higher than reported by Card although using additional explanatory variables and not using experience squared (Card, 1999).

This however means that there is a lot of wage variation that is left unexplained by the model which is likely due to omitted factors in the error term and hence omitted variable bias may be present. If unobserved factors such as ability are correlated with education and wages, education may be endogenous. An instrumental-variable approach could be considered in future research to address potential endogeneity and obtain a more credible estimate of the causal effect of education on wages (given a valid instrument can be identified).

The coefficients on tenure, marital status and number of dependents are positive, although the coefficient on number of dependents is not statistically significant at the 5% level. Therefore, there is insufficient evidence to conclude that the number of dependents is associated with hourly wages, conditional on the other variables. More importantly, the regression shows that on average men earn more than females on an hourly wage basis as the female dummy coefficient is negative and significant.

This reflects literature and can be explained by the gender pay gap investigated by Blau and Kahn (2016) which explains that by 2014, women in full-time jobs still earned only 79% of men's annual earnings, despite the gender wage gap significantly decreasing over time (Blau and Kahn, 2016).

Limitations:

Although OLS is a credited model, multiple problems can arise in estimations. Firstly, it is important to consider how data on education is collected. Card (1999) mentions problems surrounding the strong intergenerational correlation in education if parents education level is used as a measure of how educated a child is, which can lead to biased estimates. This however isn't a problem in this report as data on education is in years of education and not based on family background, but caution should be taken nevertheless (Card, 1999).

A specific problem with this data however is that it is observational data and therefore shouldn't be interpreted causally. Moreover, there are only 526 observations in this sample which is unlikely to be representative of a population and as the sample is from US in 1976, this likely has limited external validity to modern labour markets in some respects.

A major problem is omitted variable bias (OVB) in education which Card (1999) and Blundell et al (1999) highlight as a potential source of bias. Additionally, another form of bias that needs considering is measurement error which can bias coefficients downwards (Card, 1999) (Blundell et al., 1999).

Finally, although the human-capital model features a second-order polynomial for work experience, this report doesn't include it in the full model. As a result, the linear specification may not fully capture the nonlinear returns to experience. Furthermore, the experience variable represents potential rather than actual labour-market experience, and this could introduce measurement error into the estimated relationship between experience and wages.

Conclusion

Overall, this report highlights that education and work experience are positively associated with wages. As both explanatory variables are statistically significant, there is evidence of statistically significant positive associations between these variables and wages.

The full model (Regression 3) explains substantially more wage variation. The full model explains 37% of the variation in wages compared to the simple regression of wages on education and a constant where only 16.5% was explained (an increase of 20 percentage points). Although the full model explains 37% of the variation in wages, which is above the 20%-35% range reported by Card (1999) for the OLS human-capital earnings function, this comparison should be interpreted with caution as the specifications differ. For instance, additional controls such as gender and tenure were used in this report but lacked the quadratic experience term which is highlighted in the human-capital earnings function (Card, 1999).

Additionally, the full model also reveals a conditional gender wage differential, highlighting that females have an estimated hourly wage \$1.76 lower than males on average, which is indicative of the real-life gender wage gap. As literature confirms that women have hourly wages lower than men on average, despite this gap shrinking over time, these results can also be trusted.

However, caution should be taken since these results should not be interpreted causally. The main reason is due to potential endogeneity amongst explanatory variables otherwise known as the potential omitted variable bias featured particularly within the education variable highlighted by Blundell et al (1999). Therefore, instrumental variables may be a more recommended approach for this investigation in future models (Blundell et al., 1999).

Additionally, the limitations of the characteristics of this particular old and observational dataset with few observations limit its interpretability for present day applications although all directional relationships between education, work experience and wages along with the gender wage gap still hold; despite numbers being likely incorrect.

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